NOISEMAP · DESIGN DECISION RECORD

NoiseMap concept selection

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DOCUMENT CONTROL

Document ID	NSM-DDR-004	Version	0.1
Status	Draft	Date	2026-11-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/noisemap		
Source	docs/decisions/0004-concept-selection.md @ ffa26f8		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue: concept selection among options A, B and C, proposed to Amish

0004: Concept selection

- **Date:** 2026-11-02
- **Status:** Proposed, awaiting Amish. Nothing here is decided.

Context

Kit 1.8 requires at least two concept options with different working principles, scored against the user requirements. NoiseMap was designed before kit 1.8, so the current design is option A and is not redesigned without Amish's go. Options B and C were added and modelled at detail level D3 (kit 1.9).

Options

- **A, pole clamp node:** solar, own cell, LoRaWAN, microphone 4 m up on an arm (the current design).
- **B, balcony post node:** USB power, home Wi-Fi, microphone and radio on one board in a clamped box.
- **C, luminaire socket node:** twist-locks into the luminaire receptacle, takes luminaire power, LoRaWAN, microphone about 6 m up.

Scores

Weighted totals from NSM-OPT-001 Table 2 (maximum 245): A 171, B 163, C 152. A leads B by 4.7 %, which is inside the 10 % band of STANDARDS section 21, so no pick is made. A leads C by 11.1 %.

Decision requested

#	Decision	Options	Recommendation
1	Which concept is carried forward	(a) A only; (b) B only; (c) A as the street node and B as a low-cost companion for residents; (d) C	Proposed, awaiting Amish: no pick between A and B. If forced, (c) serves both user groups, but it adds a second design to maintain and changes the scope, so it is Amish's to decide. C is not recommended (mains voltage, work at height, loses the no-electrical-connection constraint)

Consequences

Until Amish decides, option A stays the design of record: the drawing, priced BOM and build plan describe A only; `concept_chosen` is "A (proposed)". If B is chosen, the BOM, build plan and calculations for B must be written (not started). If C is chosen, a mains-safety review (NSM-REG-001 G12) comes first.